PIZZA?

Ch 4.4.1: Linear Discriminant Analysis

Lecture 8 - CMSE 381

Prof. Elizabeth Munch

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Cold

Hot

Weds, Feb 2, 2022

Announcements

Last time:

- ~~Lab on Linear Regression~~

Logistic Regression

Announcements:

- Third homework due Friday! Covers:
  - Fri 1/28 Lab
  - Mon 1/31 Lecture
  - Weds 2/2 lecture ~ 4.4.1
- Office hours - ONLINE
  - This week only: No OH Thursday

Covered in this lecture

- Checking logistic questions
- Bayes theorem
- Linear Discriminant Analysis, $p = 1$

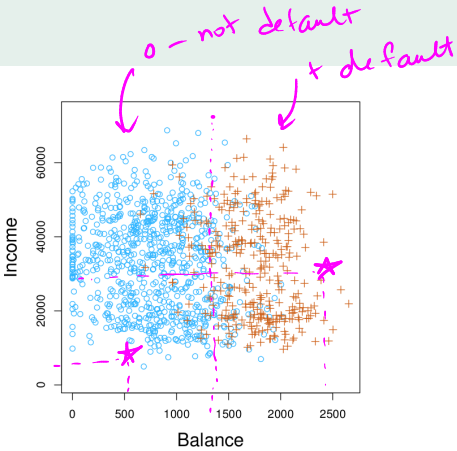
Section 1

Logistic Regression Review/Redux

What is classification

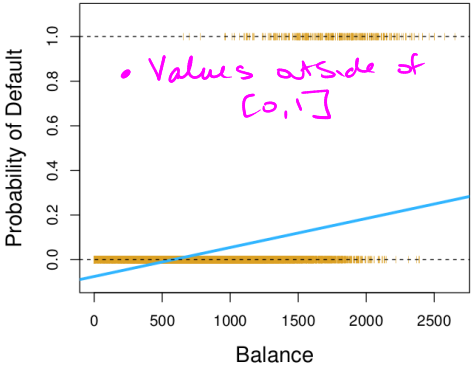
unordered split

- Classification: When the response variable is qualitative
- Goal: Model the probability that Y belongs to a particular category
- Example data:
  $p(\texttt{balance}) = \Pr(\texttt{default} = \texttt{yes} \mid \texttt{balance})$

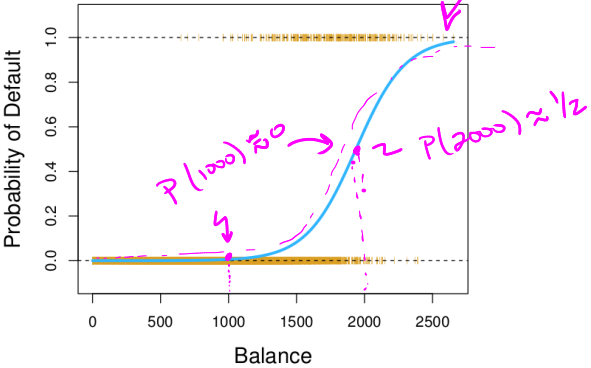


Logistic Regression

$$\Pr(\texttt{default} = \texttt{yes} \mid \texttt{balance}) = \boxed{\frac{e^{\beta_0+\beta_1\texttt{balance}}}{1+e^{\beta_0+\beta_1\texttt{balance}}}}$$



Linear Regression



Logistic Regression

Odds

probability that something happens

$Y \in \{0, 1\}$

$$\underbrace{\frac{p(x)}{1-p(x)}}_{\text{probability that something doesn't happen}} = \frac{\Pr(Y=1 \mid X=x)}{1-\Pr(Y=1 \mid X=x)} = \frac{\Pr(Y=1 \mid X=x)}{\Pr(Y=0 \mid X=x)}$$

Odds are 1/3 that someone will default

1 in 4 people will default $\quad p(x) = 0.25$

Odds $\frac{0.25}{1-0.25} = \frac{25}{75} = \frac{1}{3}$

Odds

$$\frac{p(x)}{1-p(x)} = \frac{\Pr(Y=1 \mid X=x)}{1-\Pr(Y=1 \mid X=x)} = \frac{\Pr(Y=1 \mid X=x)}{\Pr(Y=0 \mid X=x)}$$

Odds can take values from 0 to ∞

low odds/prob

high odds/prob

natural log, base e

Assume the log odds are linear

$$\log\left(\frac{p(x)}{1-p(x)}\right) = \beta_0 + \beta_1 x$$

How to get logistic function

Assume the (natural) log odds (logits) follow a linear model

$$\log\left(\frac{p(x)}{1-p(x)}\right) = \beta_0 + \beta_1 x$$

Solve for $p(x)$:

$$p(x) = \frac{e^{\beta_0+\beta_1 x}}{1+e^{\beta_0+\beta_1 x}}$$

Playing with the logistic function: https://www.desmos.com/calculator/jzsakksqcm

Interpreting the coefficients

$$p(x) = \frac{e^{\beta_0+\beta_1 x}}{1+e^{\beta_0+\beta_1 x}}$$

$$\log\left(\frac{p(x)}{1-p(x)}\right) = \beta_0 + \beta_1 x$$

increase x by 1 unit

increase log odds by 1 unit

Estimating Coefficients: Maximum Likelihood Estimation

($\mathbb{P}$ = prob Pr P)

- **Likelihood:** Probability that data is generated from a model

$$\ell(model) = \mathbb{P}[data \mid model]$$

- Find the most likely model

$$\max_{model} \ell(model)$$

- Hard to maximize likelihood, instead maximize log

$$\max_{model} \log(\ell(model))$$

- Strictly increasing log function doesn't change maximum

logistic

$$\Pr(Y = 1 \mid X) = p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

$$\ell(\beta_0, \beta_1) = \prod_{i \mid y_i = 1} p(x_i) \prod_{i' \mid y_{i'} = 0} (1 - p(x_{i'}))$$

for each data point

More on that likelihood function

$$\Pr(Y = 1 \mid X) = p(X)$$

$$= \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}}$$

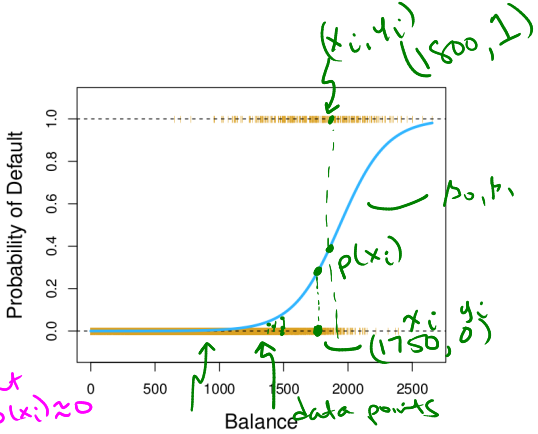
$$\ell(\beta_0, \beta_1) = \prod_{i|y_i=1} p(x_i) \prod_{i'|y_{i'}=0} \overbrace{(1-\underbrace{p(x_{i'})}_{1-0.3})}^{0.7}$$

computer max's this!

0.4

want $p(x_i) \approx 1$

want $p(x_i) \approx 0$



Multiple Logistic Regression

$X_1, \ldots, X_p \quad Y$

Multiple features:

$\exp(\beta_0 + \ldots + \beta_p X_p)$

$$P(Y=1 \mid X) = p(X) = \frac{e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}{1+e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}$$

Equivalent to:

$$\log\left(\underbrace{\frac{p(X)}{1-p(X)}}_{\text{odds}}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

Multinomial Logistic Regression

Y $K \geq 2$

What if we have a categorical variable with more than two levels (let's say K of them)?

Plan A

~~Play the dummy variable game~~:

Make K the baseline:

$$\Pr(Y = k|X = x) = \frac{e^{\beta_{k0}+\beta_{k1}x_1+\cdots+\beta_{kp}x_p}}{1 + \sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}}$$

$k = 1, \ldots, K-1$

$$\Pr(Y = K|X = x) = \frac{1}{1 + \sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}}.$$

Calculated so that log odds between two classes is linear:

$$\log\left(\frac{\Pr(Y = k \mid X = x)}{\Pr(Y = K \mid X = x)}\right) = \beta_{k0}+\beta_{k1}x_1+\cdots+\beta_{kp}x_p$$

Example

$K=3$

Predict
$Y \in \{\overset{1}{\texttt{stroke}}, \overset{2}{\texttt{overdose}}, \overset{3}{\texttt{seizure}}\}$ for
hospital visits based on X

$$\Pr(Y = \texttt{stroke} \mid X = x) = \frac{\exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x)}{1 + \exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x) + \exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x)}$$

$$\Pr(Y = \texttt{overdose} \mid X = x) = \frac{\exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x)}{1 + \exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x) + \exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x)}$$

$$\Pr(Y = \texttt{seizure} \mid X = x) = \frac{1}{1 + \exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x) + \exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x)}$$

Baseline

overdose

Softmax

$k \in \{1, 2, \ldots, K\}$

k'

Plan B: Softmax coding

Treat all levels symmetrically

$$\Pr(Y = k | X = x) = \frac{e^{\beta_{k0} + \beta_{k1} x_1 + \cdots + \beta_{kp} x_p}}{\sum_{l=1}^{K} e^{\beta_{l0} + \beta_{l1} x_1 + \cdots + \beta_{lp} x_p}}.$$

$k \in \{1, \ldots, K\}$

Calculated so that log odds between two classes is linear

$$\log\left(\frac{\Pr(Y = k | X = x)}{\Pr(Y = k' | X = x)}\right) = (\beta_{k0} - \beta_{k'0}) + (\beta_{k1} - \beta_{k'1}) x_1 + \cdots + (\beta_{kp} - \beta_{k'p}) x_p.$$

Softmax example

HW Problem

Relationship b/t the models

$$\Pr(Y = \texttt{stroke} \mid X = x)$$
$$= \frac{\exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x)}{\exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x) + \exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x) + \exp(\beta_{\texttt{seiz},0} + \beta_{\texttt{seiz},1}x)}$$

$$\Pr(Y = \texttt{overdose} \mid X = x)$$
$$= \frac{\exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x)}{\exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x) + \exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x) + \exp(\beta_{\texttt{seiz},0} + \beta_{\texttt{seiz},1}x)}$$

$$\Pr(Y = \texttt{seizure} \mid X = x)$$
$$= \frac{\left(\exp(\beta_{\texttt{seiz},0} + \beta_{\texttt{seiz},1}x)\right)}{\underbrace{\exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x)} + \underbrace{\exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x)} + \underbrace{\exp(\beta_{\texttt{seiz},0} + \beta_{\texttt{seiz},1}x)}}$$

Section 2

Generative Models

Goal:

Another way to approximate
$Pr(Y = k \mid X = x)$

How?

Bayes Theorem!

Bayes Theorem

Want to figure out

$$\overbrace{P(A \mid B)} = \frac{\overbrace{P(A)} \cdot P(B \mid A)}{\underbrace{P(B)}}$$

Posterior distribution: Prob of A being true given B is true

Prior distribution
prob of A being true

Marginalization
prob of B being true

Likelihood prob of B true given A is true

Input data

Example: Favorite language by year

Python

$X =$ freshmen

$\pi_1 \cdot P(X = \text{fresh} \mid Y = 1)$

$\frac{53}{100} \cdot \frac{9}{53} \Rightarrow \frac{9}{100}$

$\pi_1 f_1(X)$

$\pi_1 \quad \frac{53}{100}$

$+ \quad \pi_2 \quad \frac{47}{100}$

$\pi_2 f_2(X)$

$\parallel$

$P(X)$

	Frosh	Soph	Junior	Senior	
Python	9	14	13	17	53
R	14	15	10	8	47
	23	29	23	25	

$$\underbrace{P(\underbrace{Y = \texttt{py}}_{A} \mid \underbrace{X = \texttt{jr}}_{B})}_{\frac{13}{23}} = \frac{\overbrace{P(Y = \texttt{py})}^{\frac{53}{100}} \cdot \overbrace{P(X = \texttt{jr} \mid Y = \texttt{py})}^{\frac{13}{53}}}{\underbrace{P(X = \texttt{jr})}_{\frac{23}{100}}}$$

$$\frac{\cancel{53}}{\cancel{100}} \cdot \frac{13}{\cancel{53}} \cdot \frac{\cancel{100}}{23} = \frac{13}{23}$$

Following book notation

smell

large value if

high change obs w/

has X=k

Y=k

low

likelihood

- Classify an observation into one of $K \geq 2$ classes
- π_k is the *prior* probability that a randomly chosen observation comes from the kth class, $P(Y = k)$

- $f_k(X) = \Pr(X \mid Y = k)$ is the density function of X for an observation from the kth class
- $\Pr(X) = \sum_\ell \pi_\ell f_\ell(X)$

Y=l

marginalization

Bayes to the rescue!

Estimate is easy

$$\frac{\text{How many have } Y=k}{\text{How many data pts}}$$

Posterior probability that an observation $X = x$ belongs to the kth class:

$$p_k(x) = \Pr(Y = k \mid X = x) = \frac{\pi_k f_k(x)}{\sum_{\ell=1}^{K} \pi_\ell f_\ell(x)}$$

$P(Y=k)$

$P(X=x \mid Y=k)$

want to get

$P(X=x)$

estimate these

Section 3

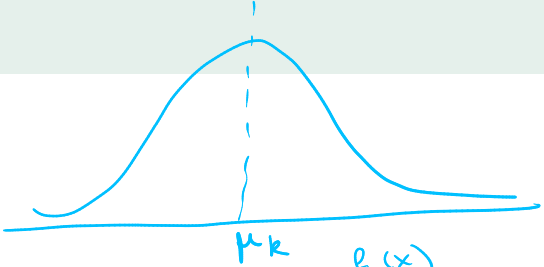
one input variable

Linear Discriminant Analysis for $p = 1$

Assumptions

Gauss

Assume $f_k(x)$ is normal/Gaussian:



$$f_k(x) = \frac{1}{\sqrt{2\pi}} \exp\left(\frac{-1}{2\sigma_k^2}(x-\mu_k)^2\right)$$

$$p_k(x) = \frac{\pi_k \overbrace{\frac{1}{\sqrt{2\pi}} \exp\left(\frac{-1}{2\sigma^2}(x-\mu_k)^2\right)}^{f_k(x)}}{\sum_{\ell=1}^{K} \pi_\ell \underbrace{\frac{1}{\sqrt{2\pi}} \exp\left(\frac{-1}{2\sigma^2}(x-\mu_\ell)^2\right)}_{f_\ell(x)}}$$

- μ_k = mean of kth class
- σ_k^2 = variance of kth class
- Assume $\sigma_1^2 = \cdots = \sigma_K^2$

$= \sigma^2$

Bayes Classifier

Same Bayes person, different Bayes definition

Bayes classifier

Assign the class k for which $p_k(x)$ is largest

Finding ~~largest~~ k st largest

$$\boxed{p_k(x)} = \frac{\pi_k \frac{1}{\sqrt{2\pi}} \exp\left(\frac{-1}{2\sigma^2}(x-\mu_k)^2\right)}{\sum_{\ell=1}^{K} \pi_\ell \frac{1}{\sqrt{2\pi}} \exp\left(\frac{-1}{2\sigma^2}(x-\mu_\ell)^2\right)}$$

([insert icky algebra])

is the same as finding ~~largest~~ k for

$$\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

largest

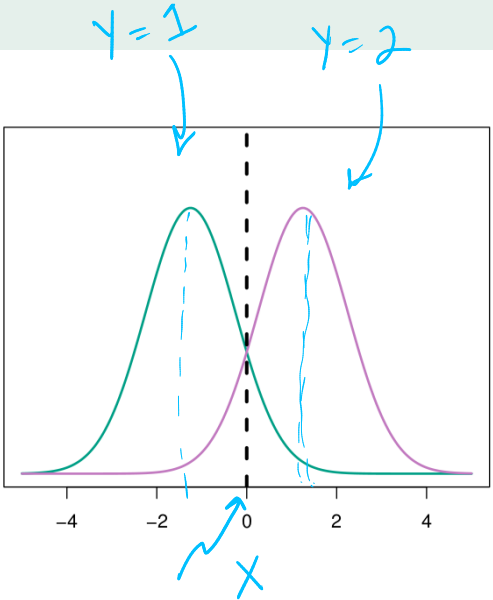
Example when $K = 2$, $\pi_1 = \pi_2$

(x_i, y_i)

Bayes

Decision boundary:

$$x = \frac{\mu_1 + \mu_2}{2}$$



New plan: Linear Discriminant Analysis (LDA)

$\delta_1(x)$ $\delta_2(x)$ Bayes decision boundary

Estimate

$$\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

- $\hat{\mu}_k = \frac{1}{n_k} \sum_{i|y_i=k} x_i$

$n_k = \# y_i = k$

- $\hat{\sigma}^2 = \frac{1}{n-K} \sum_{k=1}^{K} \sum_{i|y_i=k} (x_i - \hat{\mu}_k)^2$
- $\hat{\pi}_k = n_k / n$

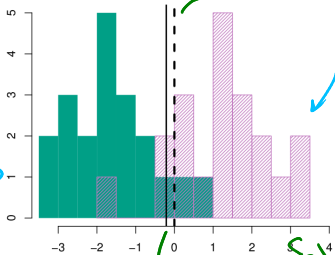
(x_i, k)

$n_k = \#\{y_i = k\}$

1 dark teal

2 = hash purple

$y=1$



(x_i, y_i)

$K=2$

Solid line: $\delta_1(x) = \delta_2(x)$

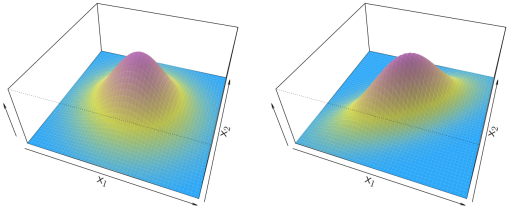
Dashed

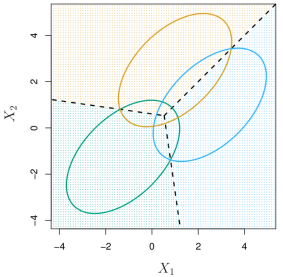
LDA review

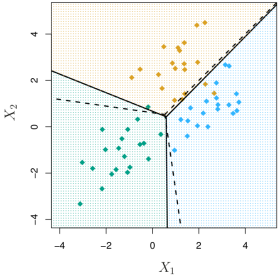
- Assume observations in each class come from normal
- Class specific means μ_k
- Common variance $\sigma_1^2 = \sigma_2^2 = \cdots = \sigma_k^2$
- Plug in estimates into Bayes classifier

Plan for next lecture (Monday)

What if you have $p > 1$?







Next time

- Thurs:
  - No office hours due to conflict, send message with issue or go to other available times!
- Fri:
  - Data ethics lab
  - HW #3 due. Covers:
    - 1/28: Linear Regression Lab
    - 1/31: 4.1-4.3 Logistic Regression
    - 2/2: 4.4.1 LDA, $p = 1$